



CS
2nd
material

Network





- 1- What term is used to describe connected computing devices that run network applications at the Internet's edge?**
- A) Routers
 - B) Switches
 - **C) Hosts**
 - D) Links
- 2- Which of the following are examples of packet switches?**
- A) Hosts and routers
 - **B) Routers and switches**
 - C) Links and hosts
 - D) Fiber and copper
- 3- What type of communication link has the highest bandwidth?**
- A) Copper
 - B) Radio
 - **C) Fiber**
 - D) Satellite
- 4- What is the primary function of routers in a network?**
- A) To store data
 - **B) To forward packets of data**
 - C) To control network access
 - D) To manage bandwidth
- 5- A network is best described as:**
- A) A single device connected to the Internet
 - **B) A collection of devices, routers, and links managed by an organization**
 - C) A protocol for sending data
 - D) A tool for running applications
- 6- The Internet is often referred to as a:**
- A) Single network
 - **B) Network of networks**
 - C) Wireless network
 - D) Local area network (LAN)



7- What connects different Internet Service Providers (ISPs) to form the Internet?

- A) Fiber optics
- B) Routers
- C) Interconnected ISPs
- D) Protocols

8- Which of the following is an example of a protocol that controls the sending and receiving of messages?

- A) HTTP
- B) TCP
- C) WiFi
- D) All of the above

9- Which of the following organizations is responsible for defining Internet standards?

- A) IETF (Internet Engineering Task Force)
- B) ISO (International Standards Organization)
- C) W3C (World Wide Web Consortium)
- D) IEEE (Institute of Electrical and Electronics Engineers)

10- What does RFC stand for in the context of Internet standards?

- A) Request for Codes
- B) Request for Comments
- C) Routing for Connectivity
- D) Resource for Communication

11- Which of the following are examples of applications supported by Internet infrastructure?

- A) Web and email
- B) Streaming video and social media
- C) E-commerce and inter-connected appliances
- D) All of the above

12- The Internet infrastructure provides a programming interface to distributed applications by offering:

- A) Hooks that allow applications to connect to Internet transport services
- B) Physical connections between devices



- C) A single protocol for all communication
- D) Hardware for data processing

13- Which of the following is an example of a human protocol?

- A) HTTP
- B) "What's the time?"
- C) TCP
- D) WiFi

14- Network protocols are primarily used for:

- A) Connecting hardware components
- B) Governing communication activity between devices on the Internet
- C) Human interaction over the web
- D) Encryption of data

15- Which of the following best defines network protocols?

- A) A set of rules for message formatting, ordering, and actions taken upon receipt
- B) A type of hardware used in routers
- C) The bandwidth of a network connection
- D) A physical connection between two devices

16- What is typically located at the network edge in the Internet structure?

- A) Routers
- B) Servers and clients
- C) Fiber optic cables
- D) Network core

17- Where are servers often housed in the network edge?

- A) In routers
- B) In data centers
- C) In wireless access points
- D) In personal computers

18- Which of the following is part of the network core?

- A) Interconnected routers
- B) Clients and servers



- C) Fiber and copper cables
- D) Wireless access points

19- Access networks primarily deal with:

- A) Data centers
- B) Wired and wireless communication links
- C) Internet service protocols
- D) Routers in the core

20- The Internet is often described as:

- A) A single network
- B) A network of data centers
- C) A network of networks
- D) A client-server architecture

21- Which of the following is a type of access network for residential connections?

- A) Institutional access networks
- B) Mobile access networks
- C) Residential access nets
- D) Fiber core networks

22- How are different channels transmitted in Frequency Division Multiplexing (FDM)?

- A) In the same frequency band
- B) In different frequency bands
- C) In alternating time slots
- D) In a single wavelength

23- What does HFC stand for in cable-based access networks?

- A) High Frequency Connection
- B) Hybrid Fiber Coax
- C) High Fiber Cable
- D) Hybrid Frequency Cable

24- In an HFC network, the downstream transmission rate can reach up to:

- A) 1 Mbps
- B) 100 Mbps



- C) 1.2 Gbps
- D) 10 Gbps

25- Which type of access networks allows mobile devices to connect to the Internet?

- A) Institutional access networks
- B) Mobile access networks
- C) Residential access nets
- D) Cable headend networks

26- What device is used to connect a telephone line to the Internet via DSL?

- A) Router
- B) DSLAM (Digital Subscriber Line Access Multiplexer)
- C) Modem
- D) Base station

27- What is the dedicated downstream transmission rate for DSL connections over a phone line?

- A) 1-10 Mbps
- B) 24-52 Mbps
- C) 10-20 Mbps
- D) 100 Mbps

28- Which of the following best describes wireless local area networks (WLANs)?

- A) They cover wide areas of up to 50 km
- B) They are typically used within or around a building
- C) They have transmission rates of up to 1 Gbps
- D) They are provided by cellular network operators

29- Which wireless standard is associated with WiFi networks?

- A) 4G/5G
- B) 802.11b/g/n
- C) 3G
- D) Bluetooth

30- What is the typical transmission rate for wide-area cellular networks such as 4G and 5G?



- A) 11 Mbps
- B) 54 Mbps
- C) 10's of Mbps
- D) 100 Mbps

31- Which of the following technologies is used for wired access in companies or universities?

- A) WiFi
- B) Ethernet
- C) 4G/5G
- D) Bluetooth

32- What is the maximum transmission rate for Ethernet wired access?

- A) 450 Mbps
- B) 10 Gbps
- C) 1 Gbps
- D) 100 Mbps

33- Which wireless technology offers transmission rates up to 450 Mbps?

- A) Bluetooth
- B) 4G
- C) WiFi (802.11b/g/n)
- D) Ethernet

34- In data center networks, high-bandwidth links can range between:

- A) 10s to 100s Mbps
- B) 10s to 100s Gbps
- C) 100s to 1000s Mbps
- D) 1 Gbps to 10 Gbps

35- What type of networks connects hundreds to thousands of servers together and to the Internet in data centers?

- A) Ethernet networks
- B) Cellular networks
- C) Data center networks
- D) Institutional access networks



36- What does a host do with application messages before sending them over the network?

- A) Encrypts them
- B) Compresses them
- C) Breaks them into smaller chunks called packets
- D) Stores them temporarily

37- In the context of data transmission, what does "R" represent?

- A) Packet length
- B) Link transmission rate (or bandwidth)
- C) Application message
- D) Packet delay

38- The formula for packet transmission delay is given by which of the following?

- A) Packet delay = $L \text{ (bits)} / R \text{ (bits/sec)}$
- B) Packet delay = $R \text{ (bits/sec)} / L \text{ (bits)}$
- C) Packet delay = $L \text{ (bits)} + R \text{ (bits/sec)}$
- D) Packet delay = $L \text{ (bits)} \times R \text{ (bits/sec)}$

39- If a packet is 2000 bits long and the transmission rate is 1000 bits per second, what is the packet transmission delay?

- A) 1 second
- B) 2 seconds
- C) 0.5 seconds
- D) 0.25 seconds

(2000 bits / 1000 bits/sec = 2 seconds)

40- What type of media allows signals to propagate freely?

- A) Guided media
- B) Unguided media
- C) Coaxial cable
- D) Twisted pair

41- Which of the following is a characteristic of twisted pair (TP) cables?

- A) They are made of glass fibers
- B) They consist of two insulated copper wires
- C) They have a bidirectional transmission capability

- D) They can only transmit signals at 10 Mbps

42- Category 6 twisted pair cables are designed for which Ethernet speed?

- A) 100 Mbps
- B) 1 Gbps
- C) 10 Gbps
- D) 100 Gbps

43- What distinguishes coaxial cables from other types of cables?

- A) They use a single copper conductor
- B) They are only unidirectional
- C) They have two concentric copper conductors
- D) They cannot carry broadband signals

44- What is a key advantage of fiber optic cables over traditional copper cables?

- A) They are cheaper to produce
- B) They have a higher susceptibility to electromagnetic noise
- C) They provide high-speed operation with low error rates
- D) They require more frequent repeaters

45- What is a characteristic of wireless radio signals?

- A) They require physical wires for transmission
- B) They can only operate in a single frequency band
- C) They can be affected by environmental factors such as reflection and obstruction
- D) They provide a full-duplex communication

46- What type of wireless connection typically offers speeds of 10-100 Mbps over short distances?

- A) Bluetooth
- B) Wireless LAN (WiFi)
- C) Terrestrial microwave
- D) Satellite

47- Which wireless technology is commonly used for mobile networks over a range of approximately 10 km?

- A) Bluetooth



- B) Wireless LAN (WiFi)
- C) 4G/5G cellular
- D) Terrestrial microwave

48- What is the maximum data rate typically achieved by terrestrial microwave links?

- A) 10 Mbps
- B) 45 Mbps
- C) 100 Mbps
- D) 270 Mbps

49- What is the primary function of packet-switching in the network core?

- A) To encrypt data
- B) To break application-layer messages into packets
- C) To store messages temporarily
- D) To increase transmission speed

50- What does forwarding (or switching) in a router refer to?

- A) Determining paths for packets
- B) Moving arriving packets from the input link to the appropriate output link
- C) Encrypting the data before transmission
- D) Increasing the data rate

51- Which of the following describes routing in the network core?

- A) Local action involving packet movement
- B) Global action determining paths taken by packets
- C) Process of encrypting data
- D) Storing packets in routers

52- What are routing algorithms primarily used for?

- A) To encrypt data
- B) To determine the best paths for packet transmission
- C) To store packets temporarily
- D) To increase transmission speeds

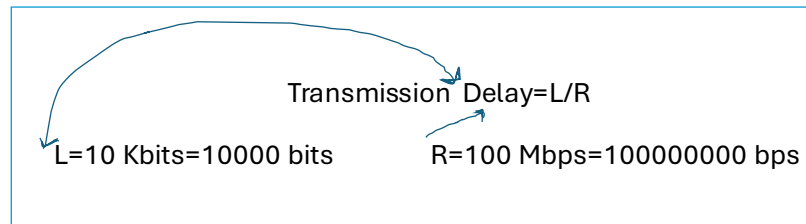
53- In packet-switching, what does "store-and-forward" mean?

- A) Only part of a packet is sent before transmission

- B) The entire packet must arrive at a router before it can be transmitted on the next link
- C) Packets are transmitted as soon as they are created
- D) Packets are permanently stored at routers

54- Given $L = 10$ Kbits and $R = 100$ Mbps, what is the one-hop transmission delay?

- A) 0.1 msec
- B) 1 msec
- C) 0.01 msec
- D) 10 msec



55- What happens when the arrival rate of packets exceeds the transmission rate of a link for some time?

- A) All packets are transmitted immediately
- B) Packets will queue, waiting to be transmitted
- C) The transmission rate increases
- D) Packets are dropped without any queuing

56- If the memory (buffer) in a router fills up, what can happen to incoming packets?

- A) They will be transmitted without delay
- B) They can be queued for processing
- C) They can be dropped (lost)
- D) They will be stored indefinitely

57- What is the main characteristic of circuit switching?

- A) Resources are shared among multiple calls
- B) End-to-end resources are allocated and reserved for each call
- C) Packets are sent without any reservation
- D) It is only used for data transmission

58- In a circuit-switched network, what happens to a circuit segment when it is not being used by a call?

- A) It can be used by another call
- B) It becomes idle and is not shared
- C) It is automatically shut down
- D) It is repurposed for data transmission

59- What does Frequency Division Multiplexing (FDM) do?

- A) Divides time into slots for calls
- B) Divides electromagnetic frequencies into narrow bands
- C) Allocates resources based on packet arrival
- D) Combines multiple signals into one

60- How does Time Division Multiplexing (TDM) allocate resources?

- A) By assigning each call its own frequency band
- B) By dividing time into slots for periodic allocation
- C) By using a shared channel for all calls
- D) By allowing simultaneous transmission of multiple calls

61- In a circuit-switched network with a 1 Gb/s link and each user requiring 100 Mb/s when active, how many users can simultaneously use the network?

- A) 5 users
- B) 10 users
- C) 15 users
- D) 20 users

$$\text{Number of Users} = \frac{\text{Total Bandwidth}}{\text{Bandwidth per User}}$$

Total Bandwidth (link speed) = 1 Gb/s = 1000 Mb/s
Bandwidth per User = 100 Mb/s

62- If each user is active only 10% of the time in a packet-switched network, what is the probability of a single user being active?

- A) 1%
- B) 10%
- C) 50%
- D) 100%

63- Why is circuit switching less efficient than packet switching in this scenario?

- A) Circuit switching requires all users to be active
- B) Circuit switching allocates dedicated bandwidth to each user regardless of their activity
- C) Packet switching guarantees performance
- D) Circuit switching can handle more users simultaneously

64- What is a major advantage of packet switching over circuit switching?

- A) Guaranteed bandwidth for every user
- B) Great for "bursty" data with resource sharing



- C) Requires no protocols for data transfer
- D) Dedicated resources for each call

65- What is a potential disadvantage of packet switching?

- A) It requires a call setup before data can be sent
- B) There is a possibility of excessive congestion leading to packet delays and loss
- C) It cannot handle bursty data effectively
- D) It always guarantees circuit-like performance

66- What is the primary function of access Internet Service Providers (ISPs)?

- A) To provide data centers for storing information
- B) To connect hosts to the Internet
- C) To manage national policies regarding Internet use
- D) To control packet transmission between routers

67- Why must access ISPs be interconnected?

- A) To reduce operational costs
- B) To ensure that any two hosts can send packets to each other
- C) To manage the flow of data to specific geographic areas
- D) To limit the number of users accessing the Internet

68- What drives the evolution of the Internet structure?

- A) Technological advancements alone
- B) User demand for higher bandwidth
- C) Economics and national policies
- D) Standardization of communication protocols

69- The Internet is often described as a "network of networks." What does this imply?

- A) It consists of multiple independent networks that are not interconnected
- B) It is a single, centralized network
- C) It consists of various interconnected networks allowing communication
- D) It operates without the need for ISPs

70- Which of the following best describes the complexity of the Internet?

- A) It is simple and easy to understand

- B) It is a well-structured, uniform network
- C) It results from various economic and policy-driven factors
- D) It operates on a single protocol

71- Occurs when memory to hold queued packets fills up?

- A) Packet transmission
- B) Packet loss
- C) Packet delay
- D) Transmission delay

72- Packets queue in?

- A) Router register
- B) Router cache
- C) Router ssd
- D) Router buffer

73- Queue length grows when arrival rate to link (Permanently) exceeds output link capacity?

- A) True
- B) False

74- $D_{\text{Nodel}} = D_{\text{Proc}} + D_{\text{Queue}} + D_{\text{Trans}} + D_{\text{Prop}}$?

- A) True
- B) False

75- Check bit errors?

- A) D_{Proc}
- B) D_{Prop}
- C) D_{Trans}
- D) D_{Queue}

76- Depends on congestion level of router?

- A) D_{Proc}
- B) D_{Prop}
- C) D_{Trans}
- D) D_{Queue}

77- Time waiting for transmission?

- A) D_{Proc}
- B) D_{Prop}
- C) D_{Trans}
- D) D_{Queue}

78- typically < microsecs?

- A) D_{Proc}
- B) D_{Prop}
- C) D_{Trans}
- D) D_{Queue}

79- D/S?

- A) D_{Proc}
- B) D_{Prop}
- C) D_{Trans}
- D) D_{Queue}

80- L/R?

- A) D_{Proc}
- B) D_{Prop}
- C) D_{Trans}
- D) D_{Queue}

81- Queuing delay small?

- A) $\frac{LA}{R} = 1$
- B) $\frac{LA}{R} < 1$
- C) $\frac{LA}{R} > 1$

82- Queuing delay large?

- A) $\frac{LA}{R} = 1$
- B) $\frac{LA}{R} < 1$
- C) $\frac{LA}{R} > 1$

83- Queuing delay infinite?

- A) $\frac{LA}{R} = 1$
- B) $\frac{LA}{R} < 1$
- C) $\frac{LA}{R} > 1$

84- Traceroute program provides delay measurement from router to source along end-end Internet path towards destination?

- A) True
- B) False

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85- Buffer has infinite capacity?

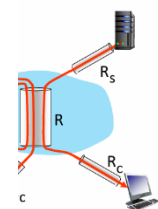
- A) True
- B) False

86- Rate at which bits are being sent from sender to receiver?

- A) Throughput
- B) Bandwidth
- C) transmission delay
- D) instantaneous

87- Choose the bottleneck?

- A) R_s, R
- B) R_s, R_c
- C) R_c
- D) R_c, R



88- promiscuous network interface records all packets passing by?

- A) True
- B) False

89- If I sent you data with another IP address then I'm doing

- A) Packet sniffing
- B) IP Spoofing
- C) DoS
- D) DDoS

90- Sending a lot of packets to the same server at the same moment called

- A) DDoS

- B) DoS
- C) Packet sniffing
- D) IP Spoofing

91- Making data secure using encryption?

- A) authentication
- B) confidentiality
- C) integrity checks
- D) access restrictions
- E) firewalls

92- Proving you are who you say you are?

- A) authentication
- B) confidentiality
- C) integrity checks
- D) access restrictions
- E) firewalls

93- reacting to DOS attacks?

- A) authentication
- B) confidentiality
- C) integrity checks
- D) access restrictions
- E) firewalls

94- digital signatures detect tampering?

- A) authentication
- B) confidentiality
- C) integrity checks
- D) access restrictions
- E) firewalls

95- Layering is important because?

- A) eases maintenance
- B) Updating one layer transparent to rest of system
- C) eases updating of system
- D) All of the above

96- supporting network applications?

- A) application
- B) transport
- C) network
- D) link
- E) physical

97- Bits “on the wire”?

- A) application
- B) transport
- C) network
- D) link
- E) physical

98- data transfer between neighboring network elements?

- A) application
- B) transport
- C) network



- D) link

- E) physical

99- routing of datagrams from source to destination?

- A) application

- B) transport

- C) network

- D) link

- E) physical

100- process-process data transfer?

- A) application

- B) transport

- C) network

- D) link

- E) physical

**101-

M

 ?**

- A) message

- B) segment

- C) datagram

- D) frame

**102-

H _l	H _n	H _t	M
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 ?**

- A) message

- B) segment

- C) datagram

- D) frame

**103-

H _n	H _t	M
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 ?**

- A) message

- B) segment

- C) datagram

- D) frame

**104-

H _t	M
----------------	---

 ?**

- A) message

- B) segment

- C) datagram

- D) frame



لا تنسو الدعاء لآخواننا في
فلسطين و غزة و السودان و
سوريا و العراق و اليمن
و تذكر دائما المقاطعة ولا
تتخلي عن مبادئك و انت بتجاهد